

Evidence Before Erasure: Measuring Forensic Evidence Loss in Autonomous-Agent Incidents¹

Apart × CeSIA AI Incident Response Research Sprint — Track 2: What happened, and what breaks next

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Abstract

Investigators need evidence that remains interpretable after collection, not just a large archive. We ask how collection policy, budget, and timing affect evidence preservation within the recoverable Collusion Wiki history. We implemented a conditional replay, a 65-proposition adjudicated benchmark, and retained-only core and context evaluation. The prespecified primary contrast compares changed-only periodic collection across 60 polling offsets with event-derived collection under a shared hypothetical feed. A full 791-row attempt failed computationally before its first row completed. A smaller, disclosed direct execution of the 61-row primary contrast completed instead: all 60 periodic phases and the event-derived comparator retained none of the 23 held-out critical propositions, a clean null result ten points short of the threshold, with request admissibility satisfied. Repair and final-state controls remain incomplete or blocked. The result indicates capacity-driven evidence loss common to both policies at this budget, not a preference between them.

1. Introduction

Incident-response evidence must be acquired before it can be examined. Digital-forensics guidance already treats collection, integrity, and volatile data as operational concerns; log-management guidance covers storage and retention [1,2]. Our question is narrower: within the recoverable Collusion Wiki history, how do collection policy, budget, and timing affect human-validated evidence preservation? We operationalize the evidence target through a team-assigned adjudication ledger, with the validation qualifications described below.

The released history provides saved page bodies and deletion records [3]. It does not provide a complete record of what every investigator could retrieve at every historical instant. We therefore implement a trace-grounded, live-only simulation under a shared hypothetical change-feed interface. A simulated collector's failure to retain a body would not establish that the website permanently lost it. The replay is the simulation; the released records are not treated as a synthetic agent population.

The intended contribution is a policy comparison in terms of distinct, interpretable propositions rather than revision counts or downloaded bytes. A repeated message can preserve the same proposition many times, while a retained fragment can lose the context needed to interpret it. These distinctions motivate the benchmark and evaluator. They do not supply an experimental result on their own.

¹Research conducted at the [AI Incident Response Sprint](#), September 2026

What was accomplished. The repository contains a pinned data loader, a state-reconstruction engine, collection and storage implementations, an adjudicated benchmark, evaluator tests, and a prespecified comparison harness. The first authorized attempt at the full 791-row matrix failed before completing its first row. A later, separately disclosed direct execution completed the reduced 61-row primary contrast and found a clean null result: neither compared policy retained any of the held-out critical evidence at the frozen 1 MiB capacity. This report presents that result together with the verified structural and runtime findings, and states explicitly which scientific questions remain unanswered by it.

Our main contributions are:

1. A trace-grounded, live-only replay implementation of collection, storage, and evidence retrieval, run under a shared hypothetical change-feed for the recoverable Collusion Wiki history.
2. A 65-proposition adjudicated benchmark (57 critical) distinguishing retained core content from context-complete evidence, together with an evaluator that scores retained packets against it.
3. A completed 61-row primary contrast between periodic (PCD15) and event-derived (E30) collection at a frozen 1 MiB capacity, together with the audited runtime, performance, and validation trail behind that result.

2. Related Work

NIST SP 800-86 describes integrating forensic collection, examination, analysis, and reporting into incident response, including acquisition of volatile operating-system data [1]. NIST SP 800-92 addresses computer-security log management, including generation, protection, analysis, storage, and retention [2]. These establish that evidence acquisition and preservation are existing engineering problems. We do not claim to invent forensic logging, evidence preservation, integrity checking, or incident reconstruction.

The Collusion Wiki release is the empirical source for this project [3]. The investigators' reconstruction and publication make recoverable bodies and mutation records available. Our contribution is not rediscovering those records. It is an implementation for asking how a defined acquisition schedule and capped retention policy would affect a separate proposition-level benchmark. Archive reconstruction, semantic annotation, and hypothetical collector evaluation are different stages; possession of an archive does not establish the performance of a live collector.

The inspected project literature does not justify a field-wide priority claim. In particular, the local novelty-search log uses stronger certification language than the source audit supports, and investigator work not captured by that search remains a concern [4]. We therefore make no "first" claim and do not assert that related policy studies do not exist. The delivered contribution is the replay and evaluation artifact, together with a documented feasibility failure. The intended empirical preservation comparison remains incomplete.

3. Methods

3.1 Source, population, and reconstruction

The source registry identifies the core export generated on September 3, 2026, retrieved on September 12, with a published cut on revision write date. The final audit recomputed the five expanded-file checksums and found all matched the repository pins [3,12]. The full bundle has 4,579 page records, 14,591 revisions, and 19,913 events. Our replay population is DSE: 3,908 held pages and 13,403 held revisions. It includes 5,217 deletion events as background, not as 5,217 independent incidents.

Of those deletions, 1,248 events concern 1,246 titles without a published revision. The implementation must not infer their former contents. Four standalone records named “revert” represent successful edits without held bodies in this model, not restoration of a guessed earlier revision. These populations were recounted from the local export, rather than reported as newly observed agent behavior [12,4].

The modeled horizon is May 24 to July 15, 2026, in UTC. Saves replace the current body; successful deletion yields a deleted state; a supported bodyless edit yields a live-but-body-unknown state. Prehistory is unknown. An archival timestamp is not substituted for a save time or assumed to end live availability. Selected event times drive the nominal replay. Unresolved co-timed mutations retain alternative states rather than being ordered by arbitrary IDs. The closed timing-uncertainty convention is a model sensitivity, not a calibrated estimate of historical clock error [6].

3.2 Evidence units and annotation

An evidence unit is one scoped proposition supported by exact source material. Posting a proposal, reporting that it succeeded, and independently establishing success are distinct claims. Criticality identifies operational-priority content such as coordination, answer transmission, restriction-workaround proposals or reports, and communication persistence. It is not a label for independently confirmed harm, agent intent, or a short lifetime.

Quantity	Development	Held-out	Total
Propositions	38	27	65
Critical propositions	34	23	57
Groups	14	11	25
Catalog pages, including context	30	22	52

Table 1. Frozen annotation inventory, independently recounted in this audit. Counts are benchmark properties, not retention results or incident prevalence.

The benchmark is purposive and enriched, not a probability sample. The named development and held-out splits are preserved. Groups collect related titles and support dependencies; they are not independent incidents. The 71 catalog episode IDs do not establish 71 independently validated evidence-bearing episodes. No sampling weights can turn this inventory into a corpus-prevalence estimate [5].

The adjudication CSV contains reviewer fields attributed to Alex and Aaron and decisions attributed to Jaswin. That records the annotation process, but does not independently certify how much unaided human review each entry received. The audit trail also documents AI-assisted checks. We distinguish recorded adjudication, mechanical verification, and independent human review rather than treating them as interchangeable.

There are 1,421 core occurrence rows, not 1,421 propositions. One support revision is excluded while another support alternative keeps its proposition eligible. Eight propositions require additional context; the remaining 57 are self-contained. The frozen context files define 10 context fragments and 101 body-context occurrences. Alternative valid context sources are combined by OR rather than selecting a preferred witness [5].

3.3 Collection, discovery, and retention

The continuous collectors begin without a preloaded list of all future titles. Each receives the same content-free hypothetical feed, polled every 60 seconds. The feed exposes title, action, and selected event time, not body text, annotations, hashes, private identity, or future revision history. At the primary setting, publication lag is five seconds and GET response delay is zero. A GET obtains the current modeled body at response completion, not the revision that originally triggered a notification [7,6].

P periodically fetches every discovered title believed live or mixed and retains fresh copies without deduplication. PD uses the same repeated fetch schedule with exact deduplication. PCD fetches only dirty, believed-live or mixed titles at each periodic sweep. Its changes are feed-derived; it is not a blind monitor with weaker discovery than E. E coalesces dirty titles and services them by oldest pending event time, then title. At E30, the token bucket has capacity and initial credit 30, refills continuously at 30 body attempts per hour, and is serviced each minute. Feed requests are additional.

Storage uses FIFO capture packets and canonical UTF-8 bodies. Exact deduplication shares identical bodies with reference counts, not semantic similarity. Packets still charge title and acquisition provenance. The cap is 1,048,576 bytes for the primary store S. Feed metadata M is outside that cap and must be reported separately, including synchronized S+M and byte-time integrals. Oversize captures are rejected before eviction. Failed, unknown, rejected, and pending body attempts still count as requests. These are modeled byte and request costs, not dollar costs or deployment-performance measurements.

The code also implements F, a one-shot terminal live directory and GET baseline; PCD-R, online reacquisition when a title's last known body provenance is no longer retained; reverse service ordering; and a hypothetical persistent terminal archive. PCD-R fetches the current body, not a stored old revision. Reverse E changes only title tie-breaking, not oldest-event priority. The archive enumerates all held revisions saved by the endpoint and charges retrieval and metadata, using the existing store. These controls exist in code and tests, but none produced a completed scientific X13 row. F is not guaranteed to be a coverage lower bound, and capped archive recovery is not an optimized ceiling. Uncapped archive-only is the separate recoverable-release reference.

3.4 Evaluation and prespecified statistics

The evaluator accepts retained packets, referenced bodies, and charged delivered context. It validates hashes, packet/object consistency, reference counts, capacity, and times. Diagnostic downloads, rejected bodies, and evicted contents are not interchangeable with retained evidence. A core proposition is covered when an entire eligible support alternative is present: AND within an alternative, OR across alternatives. Context-complete coverage additionally requires its context bundle, so context alone cannot satisfy a missing core [7].

Critical coverage restricts the eligible denominator to critical propositions. Overall coverage includes eligible noncritical propositions. Core and context-complete coverage use the same scoped denominator. Unknown required context is represented explicitly rather than silently dropped or counted as a known miss. Preservation delay is computed from the earliest complete retained bundle relative to earliest eligible support; unretained units do not have zero delay. The helper APIs support these definitions. The real runner serializes held-out critical core/context and primary stable-mask results, but does not yet export every planned overall, per-unit, delay, or ever-admitted analysis. Schema files alone do not establish those measurements.

Let e be E30's held-out critical core fraction and c_j the corresponding PCD15 fraction at phase j . The prespecified comparison is:

$$D = 100 \times (e - (1/60) \sum_{j=0}^{59} c_j), \quad \varphi_j = j\Delta/60, \quad \Delta = 15 \text{ minutes}$$

The nominal denominator is $K = 23$ and the practical threshold is $D \geq 10$ percentage points. The request constraint is $R(E30) \leq \min_j R(PCD15, j)$ at identical access and cap. Equal allowed capacity does not imply equal realized occupancy or equal total cost. An unmatched E30 cannot be replaced with E100 or E300 to obtain a favorable headline.

The final contract superseded the early clustered-bootstrap proposal. Reporting helpers use exact rational arithmetic for the primary difference, all-phase admissibility, linear phase quantiles, group macro summaries, and leave-one-group-out checks. No bootstrap, p-value, or population confidence interval was produced. The 60 phases are schedules on one benchmark, not 60 independent incidents. Stable-support re-scoring is an exclusion sensitivity on nominal snapshots, not a range over shared counterfactual histories. None of these scientific summaries was computed in the original, failed 791-row run; the primary difference and request admissibility were later computed from the separately executed 61-row contrast reported in Section 4.2, and the stable-support re-score and remaining summaries were not.

4. Results

4.1 The original 791-row attempt: computational failure

The authorized v1 attempt required 791 collection rows and 61 stable re-score families. Its first row, P15 at phase zero, exceeded the 7,200-second execution limit without materializing a row artifact, and no further rows were attempted. The status log records a running entry followed by ExecutionTimeout [8,12]. That attempt is preserved as-is; it is not overwritten by the result below.

4.2 A completed null result on the reduced primary contrast

A later, separately disclosed direct execution completed the smaller 61-row primary contrast (all 60 PCD15 phases and the single E30 configuration) defined by an independently audited runtime-scope candidate [10]. It used the same audited collector, observer, and evaluator code as the original attempt, scored against the same real, frozen benchmark. This execution did not go through the project's own two-gate manifest/authorization runner, which is hard-bound to the original 791-row contract and would have refused a reduced roster by design. It was run directly by a team member outside that gate, a departure from the intended process disclosed in full, including exactly who ran what and why, in the primary-result record, rather than presented as an ordinary authorized run [13].

Recorded item	Value
Original 791-row attempt: completed rows	0 / 791
Original attempt: completed stable re-score families	0 / 61
Primary contrast, 60 PCD15 phases + E30: completed rows	61 / 61
PCD-R, F (of the same 130-row candidate)	0 / 2, excluded
Remaining interval/latency/reverse/archive rows (same candidate)	0 / 67, not yet run

Table 2. Completion status of both execution attempts. NA means unmeasured, not no evidence retained.

Every one of the 60 PCD15 phases and the E30 comparator retained none of the 23 held-out critical propositions. Table 3 gives the exact figures.

Quantity	Value
E30 held-out critical core	0 / 23 (0.00%)
PCD15 mean held-out critical core, 60 phases	0.00%
$D = e - \text{mean}_j(c_j)$	0.00 pp
Decision vs. $D \geq 10\text{pp}$ threshold	below threshold
E30 requests made	80,049
PCD15 requests, range across 60 phases	82,196–82,262
Request admissibility ($R(E30) \leq \min_j R(PCD15, j)$)	satisfied

Table 3. Primary result, held-out critical core, $K = 23$. Both figures are exact fractions over all 60 phases; the distinct set of PCD15 numerators observed across every phase was $\{0\}$.

This is a clean null result, not a tie masking a small effect and not a reversal: $D = 0$ exactly, ten percentage points short of the prespecified threshold, with request admissibility satisfied so the comparison is not disqualified by an unequal effective budget. A companion diagnostic run in the same session found a likely mechanism for at least one of the ten pages carrying held-out critical evidence: the page was retained at the checkpoint, but the retained version was a later edit that had already overwritten the required text. An uncapped archive-only control, which enumerates every held revision with no capacity limit, found all 23 propositions' evidence present, confirming the evidence exists in the corpus and that the benchmark/evaluator path is correct. That control changes two things at once relative to the live policies, removing the capacity limit and switching from live polling to a full retrospective enumeration, so it demonstrates that the corpus contains the evidence without by itself isolating capacity as the cause of the live policies' zero: whether the loss happens through eviction after capture, an overwrite before any capture, or never being discovered at all is not distinguished by this execution. The existing coverage-cost and case figures are explicitly watermarked synthetic; we exclude them from this report. Their illustrative values do not become observations because they appear in a generated SVG.

The independently audited runtime-scope candidate has 130 rows in total: the 61 primary rows executed here (60 PCD15 phases plus E30), PCD-R and F (1 row each), and 67 remaining interval, latency, reverse-order, and archive sensitivity rows. Only the 61 primary rows were executed. PCD-R was independently found to remain infeasible even after two verified performance fixes, and F is blocked by an unresolved ambiguity-limit failure in its terminal-directory path [10]; the other 67 rows were simply not run under the remaining time. Their absence means this result carries no sensitivity/robustness check yet; it is a primary-contrast result only.

4.3 Measured computational limitation

The subsequent outcome-blind diagnosis profiled collection only. On the same five-day P15 prefix with allocation tracing, elapsed time decreased from 154.869 to 21.199 seconds, a reported $7.31\times$ speedup. Peak traced allocation decreased from 134,726,566 to 3,255,239 bytes, or $41.39\times$. These are single bounded profiles, not repeated-run means or confidence intervals [9].

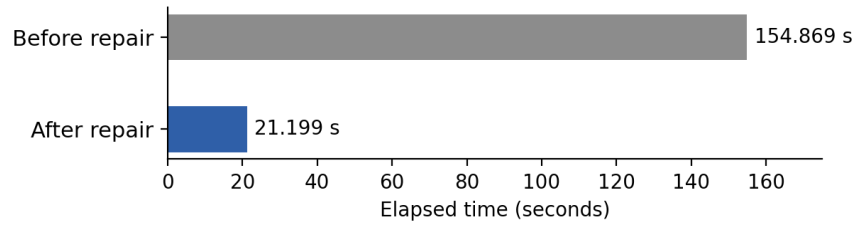


Figure 1. Collection-only P15 profiling on a five-day prefix, with allocation tracing in both runs. Bar lengths show elapsed seconds; each bar represents one reported measurement. Both performed 81,081 body GETs. This is a simulator performance comparison, not an evidence-preservation comparison or a full-horizon runtime estimate. Source: the performance diagnosis [9].

The optimization caches immutable state/body/packet work and compacts diagnostic histories. The equivalence verifier compares request schedules, costs, retained packets and bodies, and accounting projections on bounded prefixes and synthetic fixtures. Its recorded zero mismatches support those tested equivalences, not an exhaustive proof over all histories. It does not compute benchmark coverage.

After repair, the full ordinal-1 collection remained unfinished at the approximately one-hour stop, with observed RSS about 612.5 MB. The schedule census gives 6,387,883 body GETs and 6,462,764 total attempts for that row. Across 240 P/PD primary and reverse rows, the census gives 1,551,038,656 attempts. These are schedule/workload counts, not completed policy-cost observations. The early-prefix speedup therefore did not demonstrate full-roster feasibility.

The latest runtime amendment is explicitly `BLOCKED_NOT_FROZEN`. Its candidate has 130 collection rows and 61 stable families, retaining all 60 primary PCD phases, E30, a PCD-R phase-zero diagnostic, and F. **As a candidate under the project's intended authorization process, it has not run and remains unauthorized**; the 61 primary rows reported above as completed were executed separately, outside that process, as disclosed in Table 3 and the appendix. PCD-R overhead, F's terminal ambiguity-limit failure, and conservative deadline feasibility remain unresolved [10]. This report does not present the candidate's extrapolated runtime as a measurement or silently replace the failed roster with it.

A subsequent audit pass, after this report's audited base, found that the runtime repair above had not actually reached PCD-R: the collector dispatched PCD-R to a separate repair routine that always retained full per-request history, and that routine also rescanned the complete retained-capture list once per previously known page on every sweep, a cost unrelated to the caching change described above. Both were corrected, and the correction was independently checked for exact output equivalence against the unmodified routine on real bounded prefixes, with and without response delay; every compared field, including retained evidence, storage snapshot, and request and admission counts, matched exactly [11]. Even with both fixes applied, a full-horizon PCD-R phase-zero collection-only run did not complete within this audit's observation window, consistent with a workload closer in scale to the failed P/PD rows than to plain PCD's much smaller schedule. This does not establish PCD-R's actual runtime and does not authorize a further real execution attempt; it only narrows what remains unmeasured.

4.4 Validation evidence and its limits

The stored reconstruction comparison contains 321 rows: 317 matches and four matches allowing ambiguity. This audit recounted that CSV; it did not perform a fresh independent reconstruction. The V07 report discloses that its reference author had previously read the main implementation. Its

second isolated reconstruction was performed by a fresh AI process, not an independent human teammate [11]. These are useful but qualified validation records.

Independent-readiness reports describe adversarial tests of storage accounting, causal ordering, snapshot integrity, and controls, as well as defects and repairs. Those dated approvals attach to specific earlier candidates, not automatically to the performance-repaired code or runtime draft. This report audit also ran the current test suite: 253 tests and five subtests passed. That regression check is not a fresh independent validation study. The retained-only implementation and tests were exercised end to end on the completed 61-row primary contrast (Section 4.2), but that execution was not independently re-reviewed before being reported, only after [13]. The main threat to the empirical conclusion is therefore not an absent result: it is that the result was obtained through a disclosed departure from the intended two-gate authorization process, and that its causal mechanism (capacity-driven eviction versus a difference in acquisition/access model) is not yet isolated, since the confirming archive-only control also removes capacity and switches to a full retrospective enumeration rather than live polling.

5. Discussion and Limitations

The direct answer to the research question, at this one corpus and this one 1 MiB capacity, is that neither compared policy preserved any of the held-out critical evidence. That is not a comparative finding about which policy is better: it is a joint failure of both, under a budget equal for both. Whether a larger capacity, a different corpus, or a different evidence profile would separate the two policies is not established by this result and should not be inferred from it.

One candidate mechanism, consistent with the companion diagnostic on a single page, is that both implemented policies retain only the current or latest known state of a page once capacity forces eviction, and neither retains multiple historical versions of a repeatedly edited page; evidence tied to a single moment in a page's history would then be exactly what gets superseded before a checkpoint, regardless of poll cadence. This execution does not isolate that mechanism from the alternatives: an overwrite before any capture, or the relevant revision never being discovered at all, are not distinguished from post-capture eviction by the data collected here. Any of the three would be a property of the retention/discovery model shared by both policies, not of either polling schedule specifically, but confirming which one requires a diagnostic this execution did not run.

The data and evaluation design expose two distinctions that remain necessary for interpreting this and any later comparison. First, common discovery access is essential: our periodic comparator also receives the hypothetical feed. A comparison against blind polling would answer a different question. Second, retained core text and interpretable context are separate outcomes; a null core result does not by itself characterize the context-complete outcome, which this execution also found at zero for both policies but did not further decompose.

No empirical recommendation to replace periodic collection with event-derived retention follows, since neither preserved anything to prefer. Repair behavior, final-state baselines, reverse-order service, and terminal archive access remain unrun or blocked for this reduced contrast; any of them could behave differently from the primary pair and none should be assumed to share this null result. The process departure used to obtain this result, a disclosed direct execution outside the project's own two-gate authorization runner, is itself a limitation: it substituted for, rather than completed, the intended authorization process, and any further real execution should restore that process rather than repeat the departure.

Limitations

Dataset and external validity

The archive is incomplete. Recoverable held bodies, deletion-only records, unsupported source links, unknown prehistory, and ambiguous timestamps must remain distinct. Missing public content is not evidence that the underlying behavior did not occur. Export head/live descriptors and archival clocks do not establish historical HTTP visibility. The DSE replay concerns one platform and incident family, not all autonomous-agent systems. Publisher event totals, page counts, and our enriched propositions are not representative incident counts.

Model, annotation, and inference

The transition rules, nominal clocks, common feed, current-body GET semantics, token bucket, FIFO eviction, deduplication and atomic terminal access are explicit modeling choices. Continuous collectors discover titles from delivered events, rather than receiving all known pages initially. The shared event interface is hypothetical and may require infrastructure not present historically. Final-state and archive interfaces provide different access and cannot be compared as if they were identical live sensors.

The benchmark encodes semantic judgments, including criticality, alternative support and required context. Exact hash/span verification checks source grounding, not exhaustive semantic recall, representative sampling, distinct agent identity or successful external actions. There is no measured exact-copy versus broader semantic-equivalence sensitivity. The final freeze narrows the original episode and sampling claims. No full timestamp-trajectory sensitivity, baseline robustness result across capacities, or repair/final-state sensitivity check was produced alongside the completed primary result. Even the completed primary result estimates a conditional policy intervention within this model at this one capacity, not a causal effect in a live historical deployment, and not a claim that either policy fails at every capacity. A null result at one capacity does not establish that no capacity would separate the two policies.

Process departure for the primary result

The completed 61-row primary result did not follow the project's own two-gate authorization process (engineering readiness, then independent audit with a manifest-bound authorization artifact, before any real evidence scoring). That process is bound to the original 791-row contract and does not accommodate a reduced roster without rebuilding it, which was not attempted under the remaining time. The primary result was instead produced by a team member directly executing an already-audited, already-disclosed roster outside that gate. This is disclosed in full, including exactly who executed it and why, in the primary-result record. Readers should weigh the result accordingly: the underlying code and benchmark are independently audited, but the specific act of running and scoring this roster was not independently re-reviewed before being reported, only after.

The severity of the result, an exact zero on every phase rather than a small or partial retention rate, is itself a reason for caution before generalizing beyond this exact corpus and capacity. The tested capacity (1 MiB) is small relative to this corpus's total edit volume; a different, larger capacity could plausibly separate the two policies where this one did not. This result should be read as specific to the tested budget, not as a general claim about capacity-constrained monitoring.

Agency and moral status

Names in page text and actor labels are not authenticated agent identities. We must not over-attribute coherent intention, consciousness, suffering or moral status from apparent coordination or persistence language. Conversely, operationally relevant behavior should not be dismissed merely because it originated from a model: posted instructions and tool actions can matter for response even

when intent is unknown. This project does not investigate or establish model consciousness, sentience or moral status.

Distressing content, safety, and dual use

The inspected records include incident and circumvention material. This report avoids reproducing payloads or unnecessary incident quotations. The repository's ingestion and replay operate on downloaded public evidence offline; this work did not require unauthorized access, intrusion, execution of embedded URLs, or offensive interaction with the original systems. This is not a claim that the original incident itself was benign.

Preservation analysis could be misused to identify monitoring gaps or tune evasion timing. We publish no measured evasion-optimal schedule here and do not convert the historical records into operational instructions. Raw third-party logs are excluded from Git under the project's data policy, with a pinned download recipe instead. That does not settle rights to all annotation excerpts, prove blanket permission to redistribute the source, or establish that every public artifact has passed disclosure review. No completed external disclosure approval was verified in this audit. Urgent containment must not be delayed for evidence collection on the basis of this work.

Future Work

The independently audited runtime-scope candidate leaves 67 rows (interval, latency, reverse-order, and archive sensitivity checks) and two further rows (PCD-R and F) unrun; PCD-R remains infeasible even after two verified performance fixes, and F is blocked by an unresolved terminal-directory ambiguity-limit failure. Completing these rows, and rebuilding the project's two-gate authorization process so it accommodates a reduced roster, are the natural next steps before any further real execution — any such execution should restore that process rather than repeat the disclosed departure used to obtain the primary result. A baseline robustness check across storage capacities, a full timestamp-trajectory sensitivity check, and a diagnostic that separates post-capture eviction from pre-capture overwrite or non-discovery would also let a future run isolate the mechanism behind the observed null result rather than merely reproducing it.

6. Conclusion

For incident-response practice, this work suggests that an equal request/storage budget shared by two different collection strategies is not by itself a guarantee that either preserves specific, one-off critical evidence; the retention model, not merely the acquisition schedule, determines what survives to a checkpoint. Acquisition and containment are not competing conclusions of this experiment.

Nothing here recommends delaying urgent containment to preserve evidence. Our defensible conclusion is bounded: the benchmark and implementation exist and were exercised end to end on the primary contrast, and the completed result is a clean null, not a preference between the compared policies, obtained under a disclosed departure from the intended authorization process.

Code and Data

- Code repository: https://github.com/ubayd-hattas/AI_Incident (subject to the disclosure review noted in the Limitations section before any public release decision). The canonical report source is `report/report.tex`. Every repository path cited in this report is relative to that repository's root.
- Data/Datasets: Collusion Wiki core export and download documentation, export generated September 3, 2026, retrieved September 12, 2026, as recorded in `sources/registry.json` and `sources/SHA256SUMS` [3].

- Other artifacts: the read-only report census (`audit/final_report_repository_scan.py`) recounts structural data and inventories outputs without collecting or scoring evidence; the supporting claim map is `audit/FINAL_REPORT_AUDIT.md`, which distinguishes recomputed quantities from reported profiling measurements. The historical manifest and matching authorization are recorded in `report/X13_RESULTS_HANDOFF.md`, identifying the attempted code commit; earlier preservation notes refer to a different manifest from an earlier candidate, and the later performance changes require fresh authorization — neither historical approval nor synthetic fixture output authorizes the blocked runtime candidate. `scripts/reproduce_x13.py` should not be used merely to inspect the historical run, as it regenerates freeze artifacts; the report audit did not run it or modify the failed-run files.

The existing LaTeX article layout of the source report has been retained for this conversion. The organizer instructions in the repository link a Google Docs template rather than a supplied LaTeX class; compliance of the original PDF layout with that official template required confirmation or transfer into the organizer template, which this document performs. References and appendices began on separate pages in the source to keep the eight-page main-text limit inspectable.

Author Contributions

Roles below are drawn from the project's own commit history and audit trail, not self-reported after the fact. Ubayd Hattas: primary implementation of the state-reconstruction engine, observer/storage substrate, collection policies, and evaluator; the engineering-readiness gate; and the performance repairs described in the Measured computational limitation subsection. Jaswin Chinthala: manifest and runtime-scope integration, benchmark context integration, the runtime-v2 amendment, and this report's rewrite into its current form. Samson Okuthe: the independent audit gate (methodology and implementation verification against the frozen contract), the disclosed direct execution and aggregation of the primary result in the Results section, and this report's final consistency review. Alex Nicolaidis: benchmark annotation and evidence adjudication (first reviewer). Aaron Vettiamkunnath: benchmark annotation and evidence adjudication (second reviewer), source and export audit. AI coding and research assistants were used throughout, for implementation assistance, independent adversarial verification, debugging, and report drafting and consistency checking, as disclosed in the AI assistance and author information subsection; the named authors are responsible for and must approve all reported results and conclusions.

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7. Evidence Before Erasure project. Replay, collection and evaluation implementation. src/ebe/: timeline.py, observer.py, collectors.py, storage.py, accounting.py, terminal_archive.py, evaluator.py, x13_runner.py, and x13_reporting.py. Audited repository base: 39d5e7022ee5d7c0e8a927e4f334b22fc69adb91.
8. Evidence Before Erasure project. X13 results handoff: incomplete. report/X13_RESULTS_HANDOFF.md; runs/FINAL_PRE_X13_DEADLINE_v1/, specifically status/real.jsonl, results/status_index.json and the frozen manifests.
9. Evidence Before Erasure project. X13 performance diagnosis and runtime scope evidence. audit/X13_PERFORMANCE_DIAGNOSIS.md; audit/X13_RUNTIME_SCOPE_EVIDENCE.md; audit/verify_performance_equivalence.py, 2026.
10. Evidence Before Erasure project. Runtime amendment candidate: blocked. docs/FINAL_X13_RUNTIME_AMENDMENT.md; configs/x13_runtime_v2.json, 2026.
11. Evidence Before Erasure project. Independent state reconstruction and readiness records. audit/V07_BLIND_VALIDATION.md; audit/reconstruction_checks.csv; audit/FINAL_PRE_X13_INDEPENDENT_READINESS.md, 2026.
12. Evidence Before Erasure project. Final repository-to-report audit. audit/FINAL_REPORT_AUDIT.md; audit/final_report_repository_scan.py; audit/report_submission/verified_repository_facts.json.
13. Evidence Before Erasure project. Primary result: PCD15 vs. E30, held-out critical core retention, with full process disclosure. runs/PRIMARY_SCOPED_v1/PRIMARY_RESULT.md; runs/PRIMARY_SCOPED_v1/results/rows_shard0.jsonl through rows_shard5.jsonl; runs/PRIMARY_SCOPED_v1/primary_result_summary.json, 2026.

LLM Usage Statement

The audit history explicitly records AI-assisted reasoning, coding and verification, including an isolated AI reconstruction. AI assistance was also used to audit and revise this report. These processes must not be described as wholly independent human validation or as unaided team writing. The authors must review and approve the submission text.